

Notes: Ganong & Noel (AER, 2020)
**Liquidity versus Wealth in Household Debt Obligations: Evidence from
Housing Policy in the Great Recession**

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1 Big picture

- **Question.** Does mortgage policy work mainly by increasing borrower wealth through long-term debt forgiveness, or by increasing borrower liquidity through lower short-term payments?
- **Core distinction.** Principal reduction lowers long-term obligations and raises net housing wealth. Payment reduction lowers short-term cash obligations and raises disposable liquidity.
- **Main result.** Principal reduction has essentially no effect on default or short-run spending, while payment reduction substantially reduces default.
- **Mechanism.** Underwater borrowers cannot monetize extra housing wealth if principal reduction still leaves them underwater.
- **Policy takeaway.** Maturity extension and other tools that lower current payments can deliver more relief per dollar of fiscal or lender cost than principal forgiveness.

2 Incremental contribution of the paper

- **Liquidity versus wealth decomposition.** The paper isolates a wealth-only intervention and a liquidity-only intervention in real mortgage modification programs.
- **Two quasi-experiments.** The principal-reduction RD changes long-term debt while holding short-term payments fixed. The payment-reduction RD changes short-term payments while leaving NPV of obligations approximately unchanged.
- **Administrative outcomes.** The paper links HAMP, TransUnion, and JPMorgan Chase Institute data to measure default, credit card spending, and auto purchases.
- **Contribution to housing wealth MPC research.** It estimates the spending response to a wealth gain that does not relax current liquidity or collateral constraints.
- **Policy design.** It shows that debt restructurings should be evaluated by the timing of relief, not just the amount forgiven.

3 Data and measurement

3.1 HAMP and modification types

HAMP modified mortgages for distressed borrowers by targeting payments to 31 percent of income. Standard payment-reduction modifications lowered interest rates, extended maturity, and used forbearance. The Principal Reduction Alternative forgave part of unpaid balance, generating larger wealth gains while keeping near-term payments similar in the authors' comparison.

Interpretation: The program structure separates balance-sheet wealth from immediate liquidity.

3.2 Matched HAMP credit bureau file

The authors combine Treasury HAMP data with TransUnion credit bureau records. The data include loan terms, borrower characteristics, default histories, credit card spending proxies, and auto-loan-based durable consumption proxies.

Interpretation: This dataset is central for the principal-reduction analysis because it measures default and consumption around modification.

3.3 JPMCI bank dataset

The JPMorgan Chase Institute data contain de-identified mortgage and credit card account data for Chase customers from 2009 to 2016. The paper uses these data to validate consumption results and to compare private and government modifications around the 31 percent PTI cutoff.

Interpretation: The JPMCI data provide a cleaner spending measure and a setting for identifying the payment-reduction effect.

3.4 Outcomes and financial treatment variables

Default is 90-day delinquency. Consumption outcomes include credit card expenditure and auto expenditure. Liquidity is measured by immediate payment reduction. Wealth is measured by principal balance reduction and NPV reduction in mortgage payments owed.

Interpretation: The key empirical task is mapping modification features into economic margins: liquidity versus wealth.

4 Empirical specifications and model interpretation

4.1 Principal reduction RD: wealth holding liquidity fixed

$$\tau_{PR} = \frac{\lim_{v \downarrow 0} E[Y|V = v] - \lim_{v \uparrow 0} E[Y|V = v]}{\lim_{v \downarrow 0} E[PR|V = v] - \lim_{v \uparrow 0} E[PR|V = v]}.$$

Interpretation: This fuzzy RD estimates the local effect of being induced into principal reduction by the HAMP NPV cutoff.

4.2 Difference-in-differences for consumption

$$y_{i,g,t,s} = \gamma_g + \gamma_t + \gamma_{m(i),s} + \beta(PrincipalReduction_g \times Post_t) + x'_{it}\delta + \varepsilon_{i,g,t,s}.$$

Interpretation: The coefficient β is the change in spending for the principal-reduction group relative to payment-only borrowers after modification.

4.3 Payment reduction RD: liquidity holding wealth fixed

$$\tau_{pay} = \frac{\lim_{v \downarrow 31\%} E[Y|V = v] - \lim_{v \uparrow 31\%} E[Y|V = v]}{\lim_{v \downarrow 31\%} E[\Delta Pay|V = v] - \lim_{v \uparrow 31\%} E[\Delta Pay|V = v]}.$$

Interpretation: This estimates the effect of an additional percentage point of payment reduction at the HAMP PTI eligibility cutoff.

4.4 Model of MPC by home equity position

The model is a life-cycle consumption model with long-term mortgage debt and a collateral constraint. Housing wealth can be monetized only when the household has enough positive equity.

Interpretation: Principal reduction fails to stimulate underwater borrowers because it does not relax the current borrowing constraint.

5 Identification strategy

- **Principal-reduction RD.** The HAMP NPV cutoff creates a discontinuous jump in the probability of receiving principal reduction.
- **Local comparability.** Borrowers just across the cutoff should be similar except for the induced treatment probability.
- **Consumption DiD.** Principal-reduction and payment-only borrowers show parallel pretrends in credit card spending.
- **Payment-reduction RD.** The 31 percent PTI cutoff changes eligibility for HAMP and creates a discontinuity in payment reductions.
- **Interpretation.** The two designs separately identify wealth relief and liquidity relief.
- **Threats.** Estimates are local; consumption is measured with proxies; long-run effects of principal reduction are not observed; policy counterfactuals require extrapolation.

6 Tables and figures

6.1 Figure 1: Financial Impact of Modifications with and without Principal Reduction

Read: Both groups get similar short-term payment reductions, but the treatment group receives much larger balance and NPV reductions.

Economic message: This is the wealth-without-liquidity experiment.

6.2 Figure 2: Effect of Principal Reduction on Default

Read: The first stage is about 41 percentage points. The IV default effect is 0.0120 with standard error 0.0318.

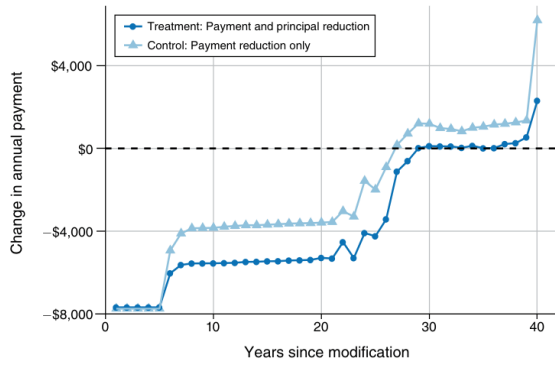
Economic message: Principal forgiveness does not reduce default when near-term payments are unchanged.

6.3 Table 1: Representativeness

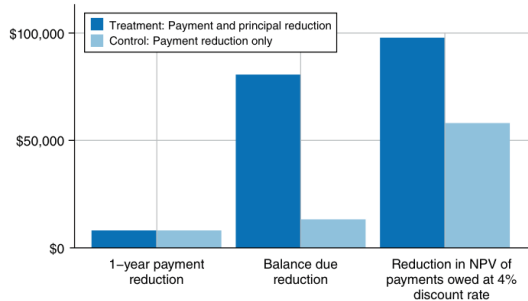
Read: The RD samples are distressed and highly leveraged, with mean LTV around 128–129 and thousands of observations.

Economic message: The samples are relevant to the policy population of distressed mortgage borrowers.

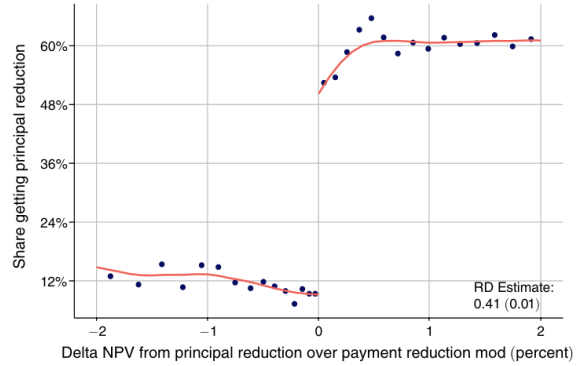
Panel A. Annual impacts on payments



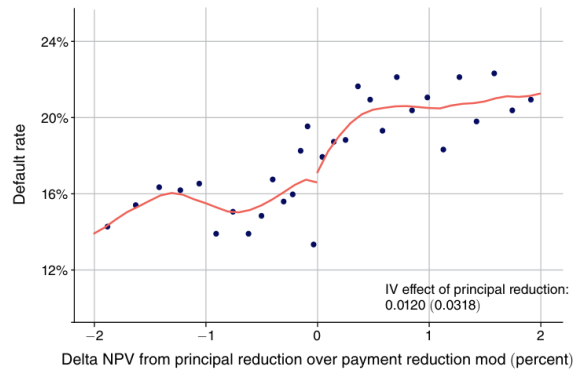
Panel B. Summary impact



Panel A. First stage: receive principal reduction



Panel B. Reduced form: mortgage default



	RD analysis sample				PSID delinquent households			
	Mean	p10	p50	p90	Mean	p10	p50	p90
Panel A. Principal reduction regression discontinuity sample								
Income	58,938	28,930	56,416	97,069	64,000	21,000	55,000	120,000
Home value	257,983	100,000	240,000	440,000	190,000	50,000	140,000	350,000
Loan to value ratio	128	104	121	167	101	52	94	166
Monthly mortgage payment	1,843	900	1,700	3,000	1,349	459	1,100	2,528
Mortgage interest rate	0.058	0.030	0.060	0.080	0.058	0.000	0.060	0.090
Mortgage term remaining (years)	26.0	23.0	25.0	34.5	23.1	10.0	25.0	30.0
Months past due	8.6	0.0	6.0	21.0	5.0	2.0	3.0	11.5
Male (<i>d</i>)	0.60	0.00	1.00	1.00	0.68	0.00	1.00	1.00
Age	48.6	36.0	46.0	66.0	43.2	31.0	42.5	57.0
Value of liquid assets					3,238	0	250	5,000
Observations	9,725				190			
Panel B. Payment reduction regression discontinuity sample								
Income	67,811	27,076	54,623	125,094	64,000	21,000	55,000	120,000
Home value	190,341	49,620	140,000	400,000	190,000	50,000	140,000	350,000
Loan to value ratio	129	63	107	205	101	52	94	166
Monthly mortgage payment	1,327	496	1,055	2,567	1,349	459	1,100	2,528
Mortgage interest rate	0.068	0.050	0.066	0.092	0.058	0.000	0.060	0.090
Mortgage term remaining (years)	22.5	15.0	24.0	26.5	23.1	10.0	25.0	30.0
Months past due	9.1	1.0	7.0	24.0	5.0	2.0	3.0	11.5
Observations	12,939				190			

Notes: This table compares borrowers in our regression discontinuity samples to delinquent borrowers in the 2009 and 2011 Panel Study of Income Dynamics (PSID) Supplements on Housing, Mortgage Distress, and Wealth Data as reported in Gerardi et al. (2015). The principal reduction sample includes borrowers with v within 0.61 percent of the cutoff (from equation (1)) and the payment reduction sample includes borrowers with PTI within 6 percent of the cutoff. All values are before modification. Panel B does not include gender, age, or liquid assets since these are not observed for this sample. The PSID sample includes heads of households who are mortgagors, ages 24–65, are labor force participants, and are 60 or more days late on their mortgage as of the survey date. The summary statistics are reported in panel A and panel B. Liquid assets include checking and savings account balances, money

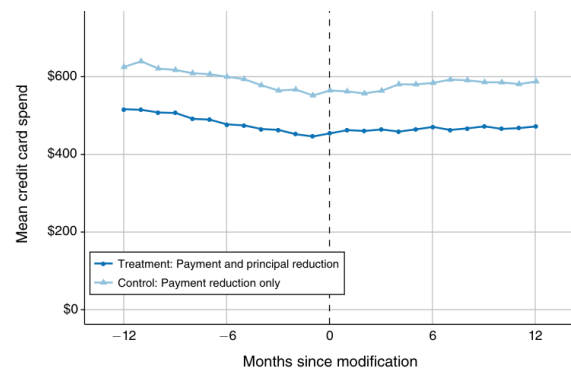


FIGURE 3. EFFECT OF PRINCIPAL REDUCTION ON CONSUMPTION

is figure empirically evaluates the impact of principal forgiveness on consumption. It shows the monthly credit card expenditure around modification for borrowers receiving each type of modified HAMP credit bureau dataset. See Section III for details.

6.4 Figure 3: Effect of Principal Reduction on Consumption

Read: Spending trends are parallel before modification and show no post-modification jump for principal-reduction borrowers.

Economic message: Balance-sheet wealth does not stimulate current spending for underwater borrowers.

6.5 Table 2: Impact of Principal Reduction on Expenditure

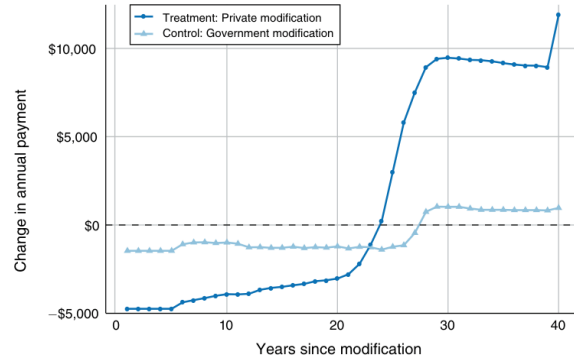
Read: Preferred estimates imply about 2 higher monthly credit card spending and 11 higher monthly auto spending, both statistically insignificant.

Economic message: The spending response to principal forgiveness is economically small.

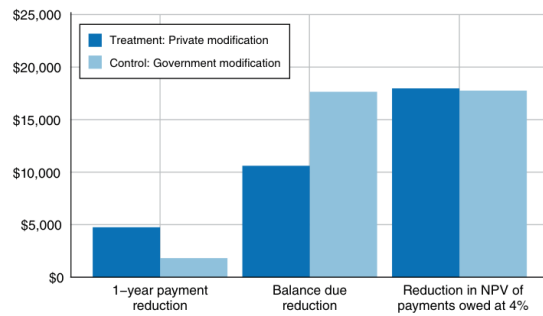
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Credit card expenditure (\$/month)</i>						
Treatment (Principal reduction × post)	0.686 (3.621)	0.721 (3.619)	0.811 (3.685)	2.068 (3.855)	0.496 (3.887)	2.144 (3.913)
MSA fixed effects		Yes				
Calendar month fixed effects			Yes			
MSA by calendar month fixed effects				Yes	Yes	Yes
Controls					Yes	Yes
Controls × post interactions					Yes	Yes
Dependent variable mean	483.82	483.82	483.82	483.82	485.55	485.55
Observations	1,678,612	1,678,612	1,678,612	1,678,612	1,642,328	1,642,328
Adjusted R ²	0.003	0.018	0.005	0.015	0.081	0.081
<i>Panel B. Auto expenditure (\$/month)</i>						
Treatment (Principal reduction × post)	13.541 (8.824)	13.545 (8.824)	13.062 (8.896)	14.648 (9.011)	15.327 (9.135)	10.994 (9.312)
MSA fixed effects		Yes				
Calendar month fixed effects			Yes			
MSA by calendar month fixed effects				Yes	Yes	Yes
Controls					Yes	Yes
Controls × post interactions					Yes	Yes
Dependent variable mean	185.84	185.84	185.84	185.84	186.62	186.62
Observations	1,678,612	1,678,612	1,678,612	1,678,612	1,642,328	1,642,328
Adjusted R ²	0.001	0.001	0.001	0.003	0.004	0.004

Notes: This table reports difference-in-differences estimates of the effect of principal reduction on expenditure in the matched HAMP credit bureau dataset. The dependent variable in panel A is monthly credit card expenditure, while the dependent variable in panel B is monthly auto expenditure computed based on balances of new auto loans. The coefficient of interest, *Treatment*, is the estimated change in the difference between outcomes of mortgages receiving modifications with and without principal reduction during the year after modification. All specifications include fixed effects for modification type and months since modification. Controls include the predicted gain to lenders of providing principal reduction, the predicted gain interacted with a dummy for this value being positive, FICO score, monthly income, pre-modification loan characteristics (LTV, principal balance, DTI, monthly pay-

Panel A. Annual impacts on payments



Panel B. Summary impacts



6.6 Figure 4: Financial Impacts of Private and Government Modifications

Read: Private modifications provide deeper near-term payment reductions, while NPV of total payments owed is similar across modification types.

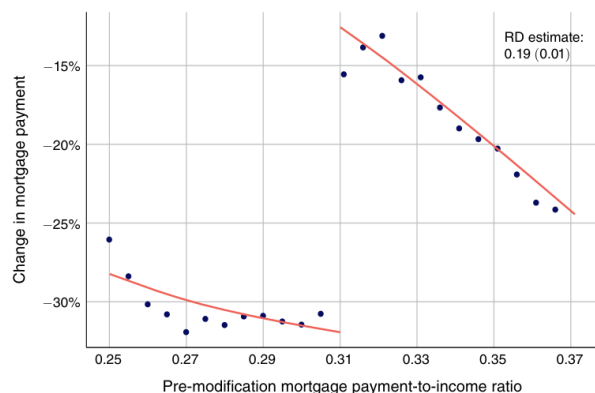
Economic message: This sets up the liquidity experiment.

6.7 Figure 5: Effect of Payment Reduction on Default

Read: Payment reductions jump by 19 percentage points at the cutoff. A 1 percent payment reduction reduces default by about 0.0038.

Economic message: Liquidity relief lowers default.

Panel A. First stage: change in mortgage payment from modification



Panel B. Reduced form: mortgage default

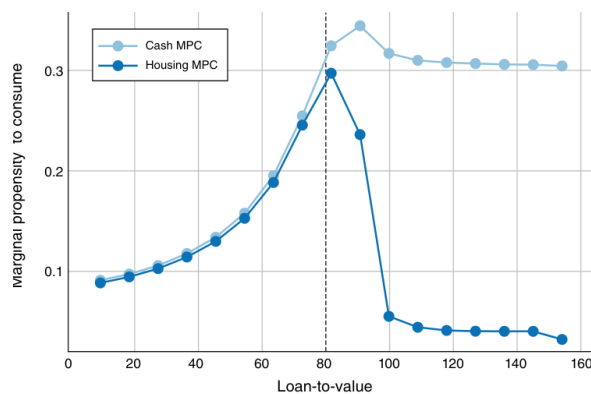
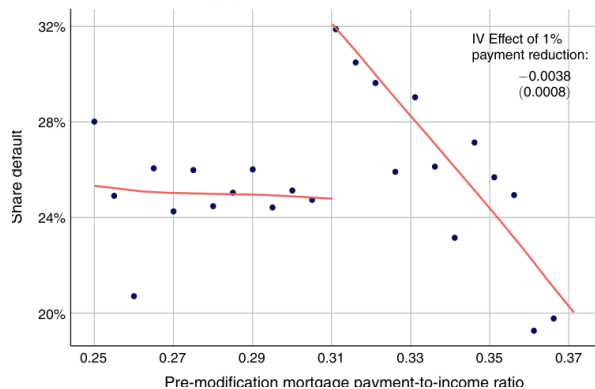


FIGURE 6. MARGINAL PROPENSITY TO CONSUME (MPC) BY HOME EQUITY POSITION IN MODEL

This figure theoretically evaluates the impact of principal forgiveness on consumption. It plots out the marginal propensity to consume (MPC) out of an additional dollar of cash or an additional wealth generated by mortgage debt forgiveness. We assume a collateral constraint of 80 percent in hand (assets + annual income) equal to 85 percent of permanent income. See Section VB for

6.8 Figure 6: Marginal Propensity to Consume by Home Equity Position in Model

Read: Cash MPC remains high for underwater borrowers, but housing-wealth MPC collapses when borrowers are far underwater.

Economic message: Housing wealth is not cash; it matters only when it relaxes collateral constraints.

7 Interpretive synthesis

Default and consumption responses are driven by current liquidity, not by long-term balance-sheet wealth. Principal reduction may matter later, but it is not an effective short-run stabilization tool for underwater borrowers who remain collateral constrained.

8 Short summary

- Principal reduction changes wealth but not short-term liquidity and has no detectable default or spending effect.
- Payment reduction changes liquidity and substantially reduces default.

- The model explains the zero housing-wealth MPC for underwater borrowers through collateral constraints.
- The policy lesson is that modifications should maximize near-term payment relief per dollar of cost.

9 Conclusion

9.1 Core conclusion

Liquidity, not long-term wealth, is the key margin for distressed mortgage borrowers in this setting.

9.2 Incremental contribution in one sentence

The paper provides quasi-experimental evidence that the timing and liquidity of debt relief matter more than the nominal size of principal forgiveness for default and short-run consumption.

9.3 Takeaway

Mortgage restructurings should be designed around the borrower constraint that is actually binding: current cash flow.